

Tasks

This experiment deals with the observation of charged oil droplets, which are accelerated between two capacitor plates.

1. Measure some rise and fall times of oil droplets at different voltages
2. Determine the radii and the charges of the droplets
3. From your results determine the elementary charge e

Remarks

It is strongly recommended to perform the experiment in several groups of at least two students each. So one can collect all results of each group at the end of the evaluation to obtain a more exact value for the elementary charge.

Duration: approx. 2 hours (the duration of the experiment depends on the numbers of observed charged oil droplets. 2 hours is for the observation of three charged droplets).

Equipment

Millikan apparatus	09070.00	1	Circular level	02122.00	1
Multi-range meter w. overl. prot.	07021.01	1	Connecting cord, $l = 100$ mm, black	07359.05	1
Power supply, 0...600 VDC	13672.93	1	Connecting cord, $l = 750$ mm, red	07362.01	3
Stage micrometer, 1 mm – 100 div	62171.19	1	Connecting cord, $l = 750$ mm, blue	07362.04	3
Stop watch, interruption type	03076.01	2	Connecting cord, $l = 750$ mm, green-yellow	07362.15	1
Cover glasses 18×18 mm, 50 pcs.	64685.00	1			
Commutator switch for Millikan experiment	06034.07	1	Optional accessories		
Tripod base -PASS-	02002.55	1	Radioactive source, Am-241, 74 kBq	09047.51	1
Stand tube	02060.00	1	FlexCam Scientific Pro II	88030.93	1
			TV set		

Setup

Set up the experiment according to the following instructions and pictures:

- Connect the capacitor of the Millikan apparatus to the commutator switch as shown in Fig. 1. Use the circular level for an horizontal alignment of the apparatus.



Fig. 1

- Use the small black connecting cord to connect the fixed (300 V d.c.) and the variable (0 to 300 V d.c.) outputs of the power supply in series (Fig. 2) and use the yellow-green connecting cord to earth the Millikan apparatus (Fig. 2 and 3)

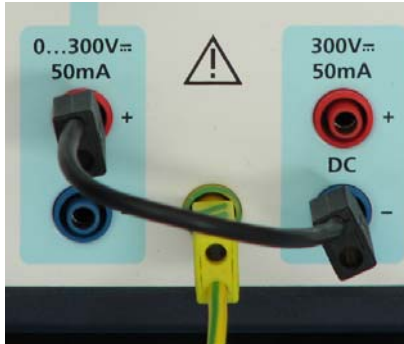


Fig. 2



Fig. 3

- Now, connect the commutator switch to the power supply (Fig. 4 and 5) and to the multi-range meter (Fig. 6)

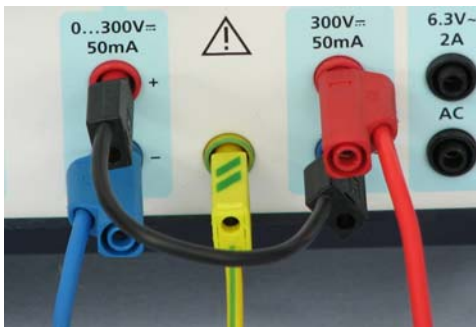


Fig. 4

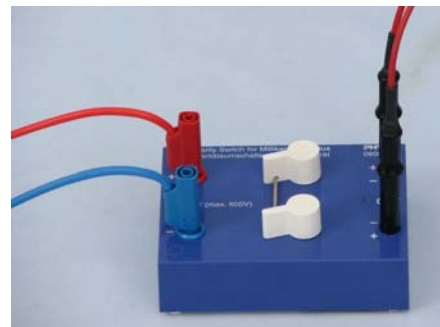


Fig. 5

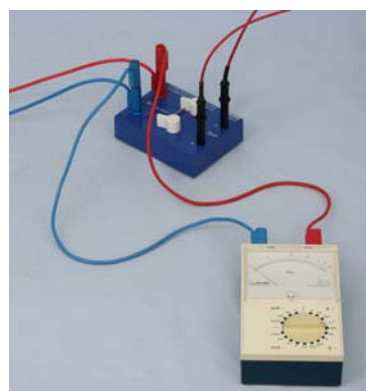


Fig. 6

- The lighting system of the Millikan apparatus is connected to the 6.3 V a.c. sockets of the power supply as shown in Fig. 7 and 8

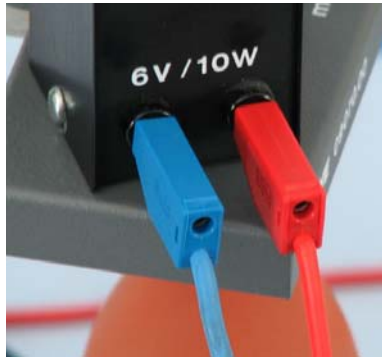


Fig. 7



Fig. 8

- Your setup should now look like the following picture:

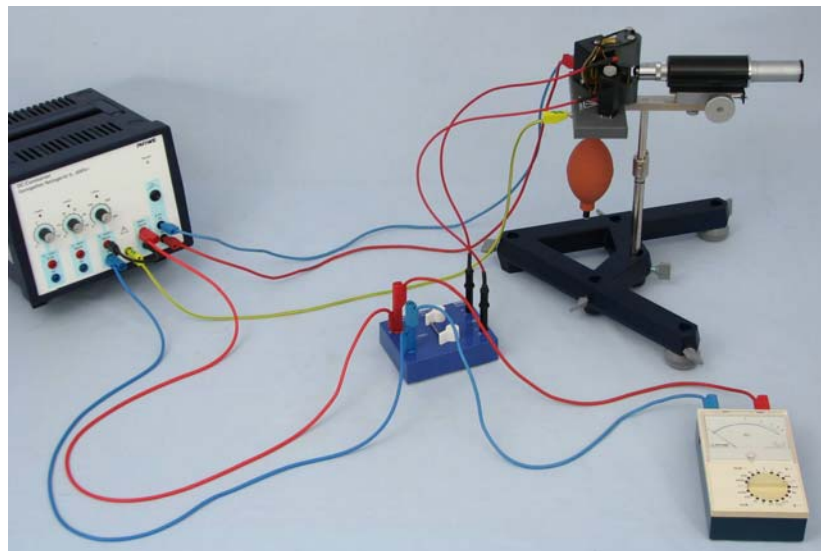


Fig. 9

Procedure

Calibration:

Calibrate the eyepiece micrometer of the microscope with the stage micrometer. In the middle of the stage micrometer there are two little circles and in the middle of the smaller one there is a scale. This scale is used to calibrate the eyepiece micrometer.

- To do this, slip the stage micrometer into the slits between the lens of the microscope and the opening of the little chamber (Fig. 10)

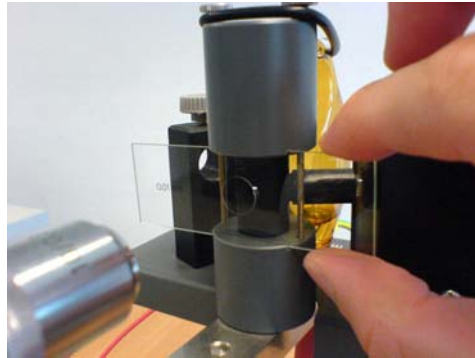


Fig.

- Look through the microscope and adjust the focussing so, that you can see the scale (the two circles around it should be used as orientation). The distance between the lens of the microscope and the stage micrometer should be only a few millimetres
- The two scales must lie one on top of the other
- Now, you can count the lines of the stage micrometer, which are needed to cover the 30 lines of the eyepiece micrometer
- From this, you can convert the 30 div of the eyepiece micrometer into mm, since 100 lines on the stage micrometer correspond to 1 mm
- Note your result on page 6
- Remove the stage micrometer from the slits to start the experiment
- Sometimes it is useful to put a cover glass into the slits to protect the chamber against air draft caused by air condition etc.

Preparation:

- Select the 600 V d.c. measurement range on the multi-range meter as shown in Fig. 11

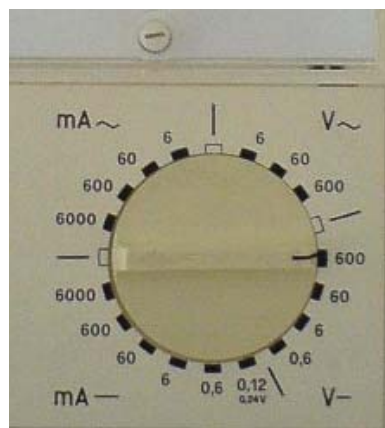


Fig. 11

- Switch on the power supply and set the capacitor voltage to 300 V (turn the rotary switch for the voltage on the power supply to 0)

- Look through the microscope and blow in some oil droplets by pressing the bellows a few times
- Adjust the focussing of the microscope until you see the oil droplets as little white circles (Fig. 12)

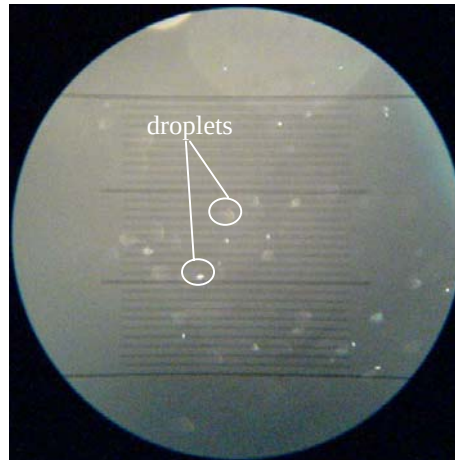


Fig. 12

- Observe the behaviour of the oil droplets and find a charged one (an oil droplet can be identified as a charged one when it changes its direction by switching the commutator switch, which inverts the polarity of the capacitor)
- To see an oil droplet for a longer time, you might have to correct the focussing of the microscope
- Try to move a charged oil droplet a few times between the highest and the lowest line on the eyepiece micrometer by switching the commutator switch
- Get a feeling for the controlling of an oil droplet
- When you do not see oil droplets anymore, blow in some droplets again (sometimes it is useful to switch off the power supply and switch it on again after a few seconds)
- For a more comfortable observation you may use the optional FlexCam in combination with a TV set

Measurements (Task 1):

- When you are ready to start the measurements, your partner should take the stop watches
- Find a charged oil droplet, which has such a velocity, that it needs about 1...3 s for the distance of 30 div on the eyepiece micrometer
- When you have found such an oil droplet, sum up some rise times with the first stop watch and some fall times with the second stop watch. Pay attention to the fact that the image, which you can see through the microscope, is upside down
- The added times should be larger than 5 s in both cases
- Note your measured rise (t_1) and fall times (t_2) in Table 1 on page 6 (columns 2 and 5)
- Record the distance s , which the oil droplet covered in these times in Table 1, too (columns 3 and 6)
- Repeat the measurement twice; once for a voltage of 400 V and another time for a voltage of 500 V
- Again, note your results in Table 1
- Usually only a few of the oil droplets are charged. You can increase their number by using the optional radioactive source

Results:

Calibration:

30 div =

Measurements:

Note your measurement results in the table below:

Table 1

Voltage U [V]	Rise times t_1 [s]	Covered distance s_1 [div]	Covered distance s_1 [mm]	Fall times t_2 [s]	Covered distance s_2 [div]	Covered distance s_2 [mm]
300						
400						
500						

With the help of your calibration, convert the covered distances of the oil droplets into mm and note your results in Table 1, too (columns 4 and 7).

Sample results

30 div = 0.89 mm

Voltage U [V]	Rise times t_1 [s]	Covered distance s_1 [div]	Covered distance s_1 [mm]	Fall times t_2 [s]	Covered distance s_2 [div]	Covered distance s_2 [mm]
300	8.1	60	1.78	6.3	60	1.78
400	8.8	60	1.78	5.2	60	1.78
500	9.8	120	3.56	5.5	90	2.67

Evaluation:

Task 2:

The values for the velocities v_1 and v_2 for each of the three oil droplets can be calculated from your measurement results s_1 and t_1 or s_2 and t_2 respectively (see Table 1) with the equation

$$v = \frac{s}{t}.$$

Do not forget to convert your results for s_1 and s_2 into the unit m. Note your results in Table 2 below:

Table 2

Rise velocity v_1 [m/s]	Fall velocity v_2 [m/s]	Charge Q [C]	Radius r [m]

Now calculate the oil droplets' charges Q as well as their radii r and note your results in Table 2 using following equations (for the derivation refer to the appendix):

$$Q = C_1 \times \frac{v_1 + v_2}{U} \sqrt{|v_1 - v_2|}$$

where

$$C_1 = 2.73 \times 10^{-11} \text{ kg (m/s)}^{-\frac{1}{2}}$$

and

$$r = C_2 \times \sqrt{|v_1 - v_2|}$$

where

$$C_2 = 6.37 \times 10^{-5} \text{ (m} \times \text{s)}^{\frac{1}{2}}.$$

Sample results

Rise velocity v_1 [m/s]	Fall velocity v_2 [m/s]	Charge Q [C]	Radius r [m]
2.20E-04	2.83E-04	3.63E-19	5.06E-07
2.02E-04	3.42E-04	4.39E-19	7.54E-07
3.63E-04	4.85E-04	5.11E-19	7.04E-07

Task 3:

In order to determine the elementary charge e from the measurement results, collect the as many results as possible for the oil droplets' charges Q and their radii r . Then draw the charges Q against the radii r of the oil droplets (you may use for this the Phywe software "measure" which you can download for free at www.phywe.com - see appendix):

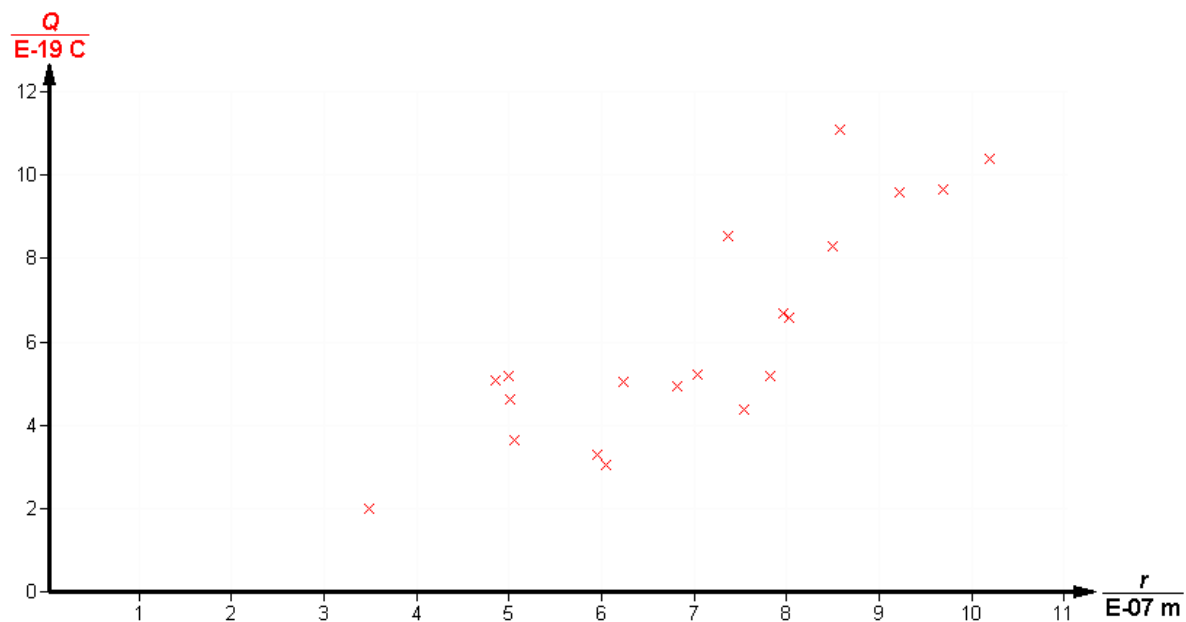


Fig. 13

Now, draw in the lines, which characterise each charge level. Your graph should then look like the following:

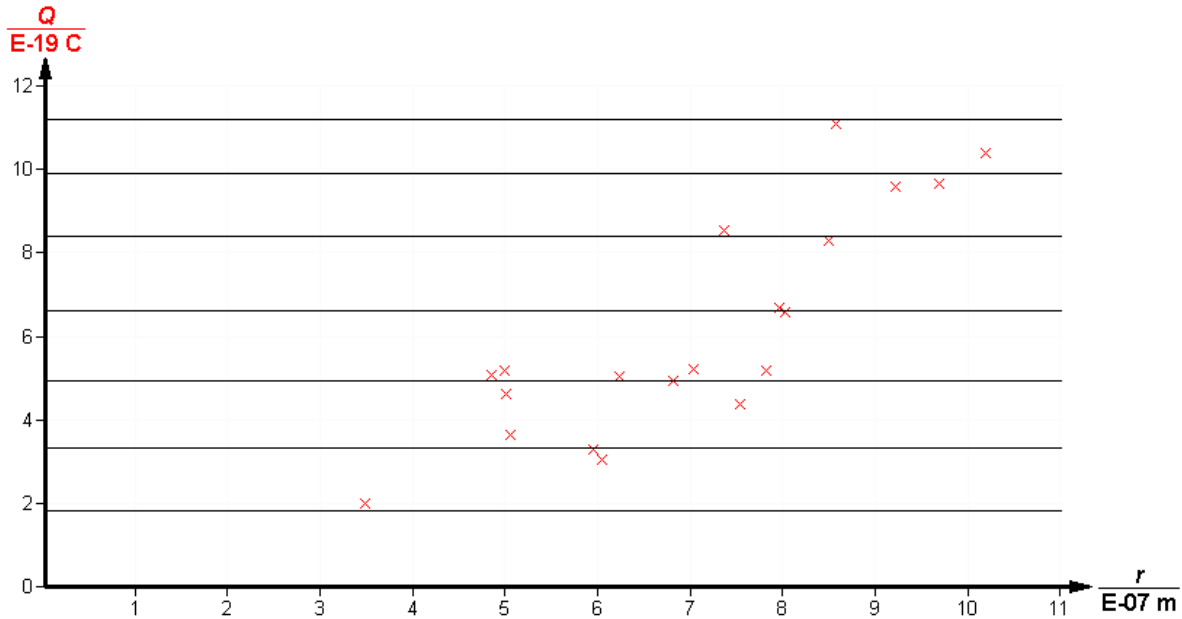


Fig. 14

In the next step, determine the distance between each line. These distances correspond to the elementary charge e . Do not forget the distance between the x-Axis ($Q = 0$) and the first line. Note down your results in the table below and calculate the mean value of the distances:

Table 3

Investigated lines	Distance [C]
x-Axis and first line	
First and second line	
Second and third line	
Third and fourth line	
Fourth and fifth line	
Fifth and sixth line	
Sixth and seventh line	
Mean value	

Sample results

Investigated lines	Distance [C]
x-Axis and first line	1.82E-19
First and second line	1.50E-19
Second and third line	1.61E-19
Third and fourth line	1.68E-19
Fourth and fifth line	1.78E-19
Fifth and sixth line	1.50E-19
Sixth and seventh line	1.29E-19
Mean value	1.60E-19

The literature value for the elementary charge e is:

$$e = 1.602 \times 10^{-19} \text{ C.}$$

Further questions

1. Explain, why the charged oil droplet rises or falls respectively.

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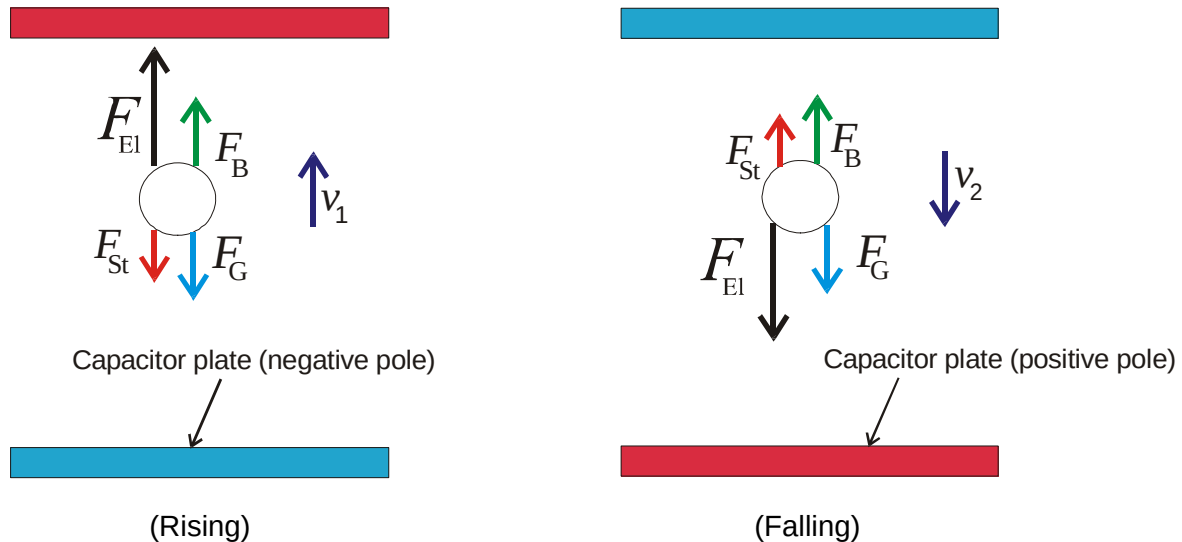
2. In this experiment, several forces take effect on the charged oil droplets. Design two graphs; use the first graph to draw in the forces, which take effect on a charged oil droplet when it is rising and use the second graph to draw in the active forces when a charged oil droplet is falling:

3. Which correlation can be observed between the charge of an oil droplet and the elementary charge? Formulate this correlation with a formula considering the oil droplet's charge Q , the elementary charge e and the number of charge levels n :

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Sample answers:

- When the oil droplets are blown in between the capacitor plates, some of them are charged by the electrical field. As we know, the charge of the oil droplets is negative. That is the reason why a charged oil droplet moves towards the positive pole of the capacitor. In our case, as it can be seen in the sketch above, it would rise at first.
By switching the commutator switch, the poles of the capacitor are inversed and the charged oil droplet falls.
- Your graphs should look like the following:



The respective forces are:

1. F_{El} = Force of the electrical field
 2. F_B = Force of buoyancy
 3. F_{St} = Force of friction (Stoke's Law)
 4. F_G = Gravitational force
- The correlation between the charge of an oil droplet Q and the elementary charge e , which can be obtained from the results of the experiment, is:

$$Q = n \times e$$

where n is the respective charge level. Since e is not divisible, n is a natural number.

Appendix

Determination of the charge and radius of an oil droplet

In order to determine the charges and the radii of the observed oil droplets, the following considerations are made:

A sphere (in our case the oil droplet) of radius r and velocity v , which moves through a medium (in this experiment air) of viscosity η , experiences the force F :

$$F = 6 \times \pi \times r \times \eta \times v \quad (\text{Stoke's law}) \quad (1)$$

where at this experiment $\eta = 1.82 \times 10^{-5} \text{ kg} \times (\text{m} \times \text{s})^{-1}$ is the **density** of air.

Since the oil droplet has a mass m (and volume V and density ρ_1 respectively), the gravitational force takes effect on it with the local gravitational acceleration g (between 9.78 and 9.82 m/s^2 depending on latitude):

$$F = m \times g = \rho_1 \times V \times g, \quad (2)$$

where $\rho_1 = 1.03 \times 10^3 \text{ kg m}^{-3}$ is the density of the oil droplet and $V = \frac{4}{3} \pi r^3$ its volume.

Furthermore, the oil droplet experiences the force of buoyancy of the air (density $\rho_2 = 1.293 \text{ kg m}^{-3}$)

$$F = \rho_2 \times V \times g \quad (3)$$

as well as the force of the electrical field inside of the capacitor

$$F = Q \times E = Q \times \frac{U}{d}, \quad (4)$$

where Q the charge of the oil droplet, U the capacitor voltage and $d = 2.5 \text{ mm} \pm 0.01 \text{ mm}$ the distance between the two capacitor plates.

From these forces acting on the charged oil droplet following rise (v_1) and fall velocities (v_2) can be obtained:

$$v_1 = \frac{1}{6 \pi r \eta} \left(Q \times \frac{U}{d} - \frac{4}{3} \pi r^2 g (\rho_1 - \rho_2) \right), \quad (5)$$

$$v_2 = \frac{1}{6 \pi r \eta} \left(Q \times \frac{U}{d} + \frac{4}{3} \pi r^2 g (\rho_1 - \rho_2) \right). \quad (6)$$

Now, one can use (5) and (6) to form the equation for the oil droplet's charge Q as well as the equation for its radius:

$$Q = C_1 \times \frac{v_1 + v_2}{U} \sqrt{|v_1 - v_2|}, \quad (7)$$

where

$$C_1 = \frac{9}{2} \pi d \times \sqrt{\frac{\eta^3}{g(\rho_1 - \rho_2)}} = 2.73 \times 10^{-11} \text{ kg (m/s)}^{1/2}$$

and

$$r = C_2 \times \sqrt{|v_1 - v_2|} \quad (8)$$

where

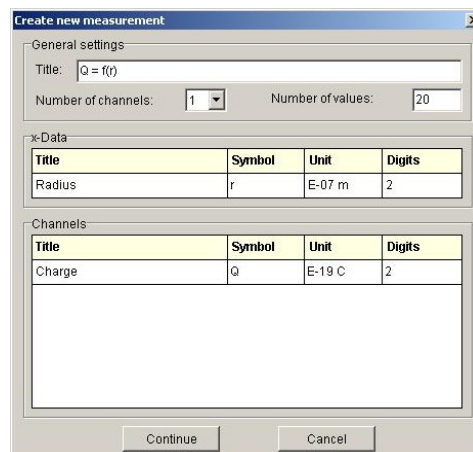
$$C_2 = \frac{3}{2} \times \sqrt{\frac{\eta}{g(\rho_1 - \rho_2)}} = 6.37 \times 10^{-5} \text{ (m} \times \text{s)}^{1/2}.$$

In order to understand, why the unit of the charge is [As], use the following considerations. If one has a look at equation (7) and only considers the units, one gets:

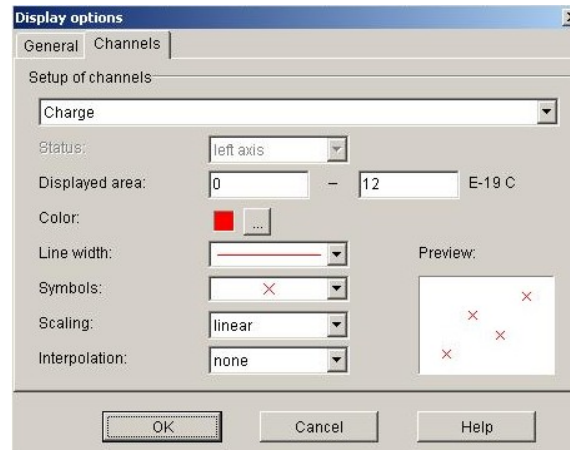
$$\text{kg} \sqrt{\frac{\text{m}}{\text{s}}} \times \frac{\left(\frac{\text{m}}{\text{s}}\right)}{\text{V}} \times \sqrt{\frac{\text{m}}{\text{s}}} = \text{kg} \frac{\text{m}}{\text{s}} \times \frac{\left(\frac{\text{m}}{\text{s}}\right)}{\left(\frac{\text{kg m}^2}{\text{s}^3 \text{ A}}\right)} = \text{kg} \frac{\text{m}^2}{\text{s}^2} \times \frac{\text{s}^3 \text{ A}}{\text{kg m}^2} = \text{A s} = \text{C}$$

Using the software "measure" for plotting and evaluating the data

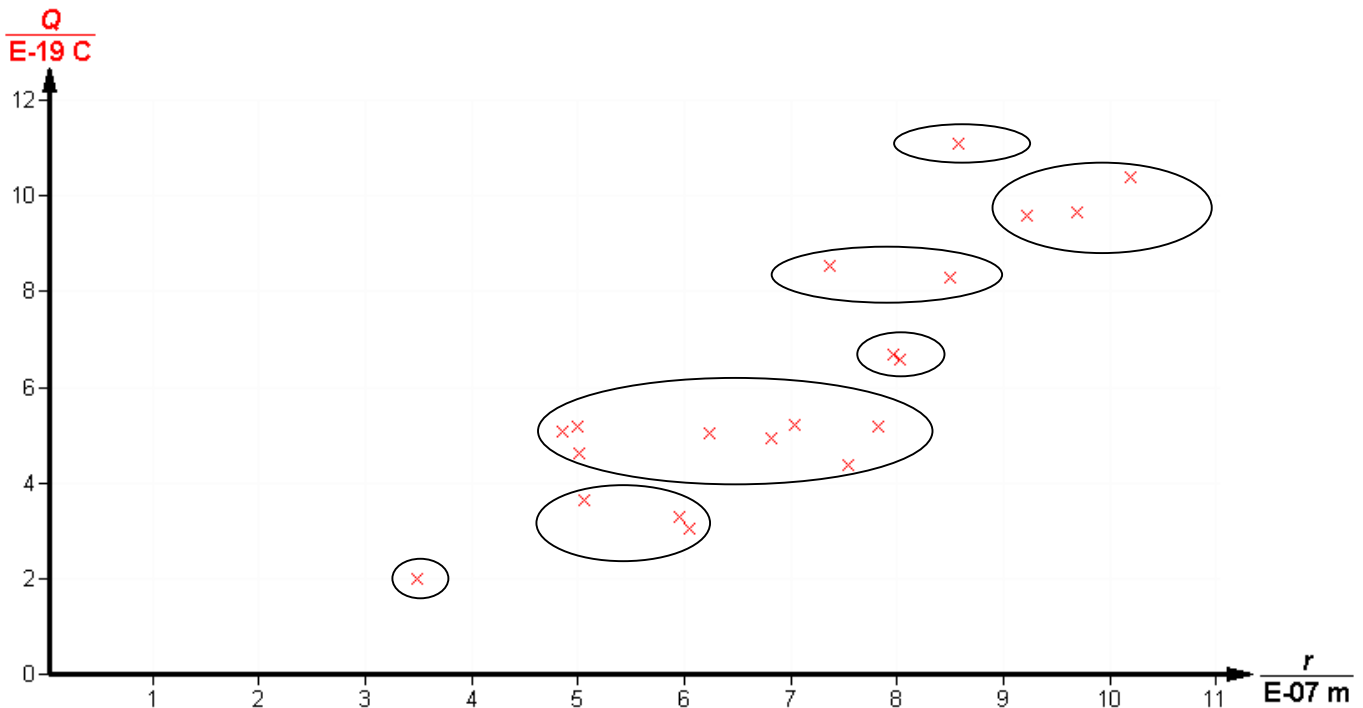
- Once installed, start "measure" and click "Measurement" and choose "Enter data manually".
- There, select the parameters that are shown in the screenshot below (your "Number of values" may be different):



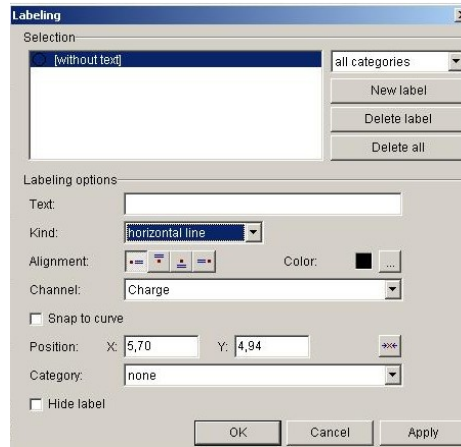
- Click "Continue"
- Type in your values for r and Q (make sure to use commas instead of points for decimal numbers)
- Click "OK"
- In the appearing dialogue box choose the first point "Sort x-data..."
- Right-click on your graph and select "Display options" (Symbol: )
- Under "channels" select "interpolation" and "none" for displaying the points only. In this menu you can also change the displayed area for the best display



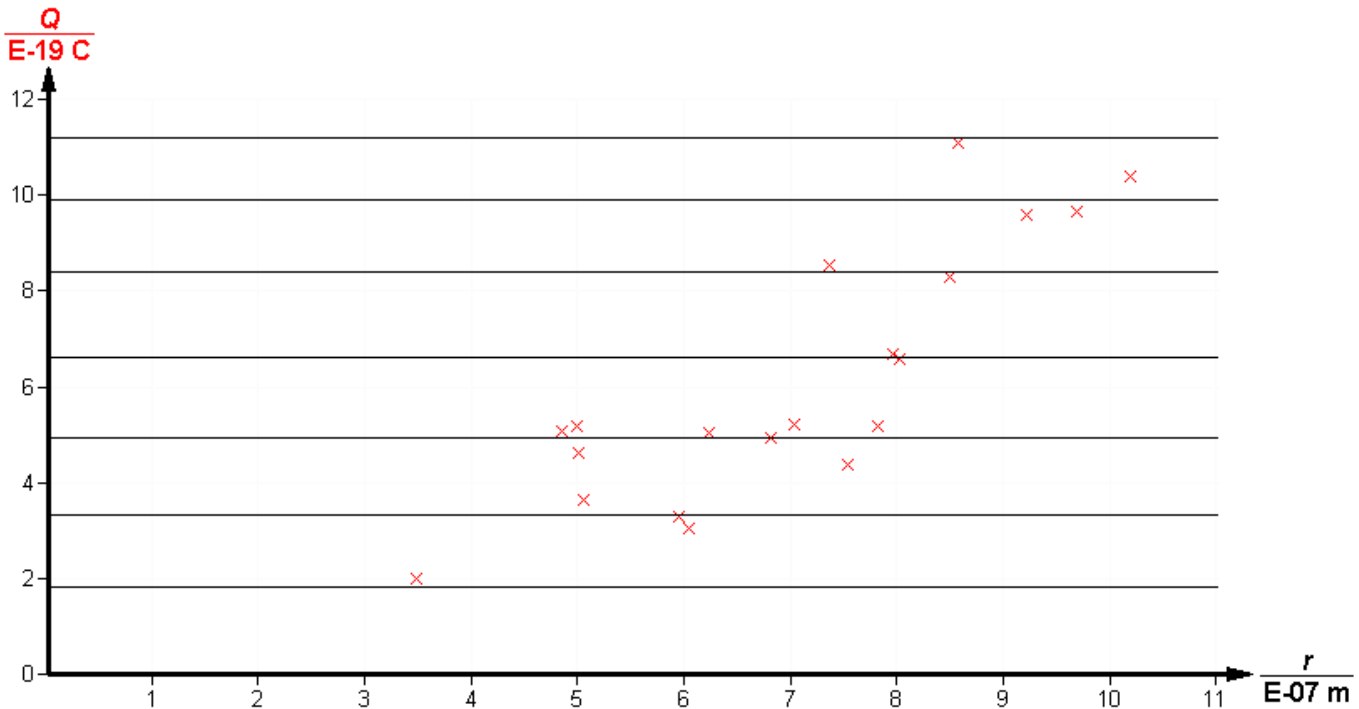
- When you look at the points, you should recognise, that there is no continuous distribution but that the points lie on different charge levels



- Now, draw in the lines, which characterise each charge level
- Click the "Label" button (A)
- Place the cross that you can see now in the middle of each charge level, e.g. try to place it in such a way, that approx. the same number of points are above and under the centre of the cross (like a kind of "manual fitting")
- Click the left mouse button, select under "Kind" "horizontal line" and click "OK"



- Now, you will see a line, which indicates a charge level
- In this way, draw in the lines for the other levels, too
- Your graph should then look like the following:



- In the next step, determine the distance between each line. These distances correspond to the elementary charge e
- To do this, click the "Survey" button ()
- You will now see two points labelled with "1" and "2"
- These two points can be moved. To determine the distance between two lines, move the point "1" on one line and the point "2" on the next line
- The distance is given with the value for ΔY as shown in the following screenshot

Elementary charge and Millikan experiment

Students' worksheet

LEP
5.1.01-00

